# il2cpp

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# 목차

1. il2cpp

2. Tool Tool Tool..

3. Game Analyze

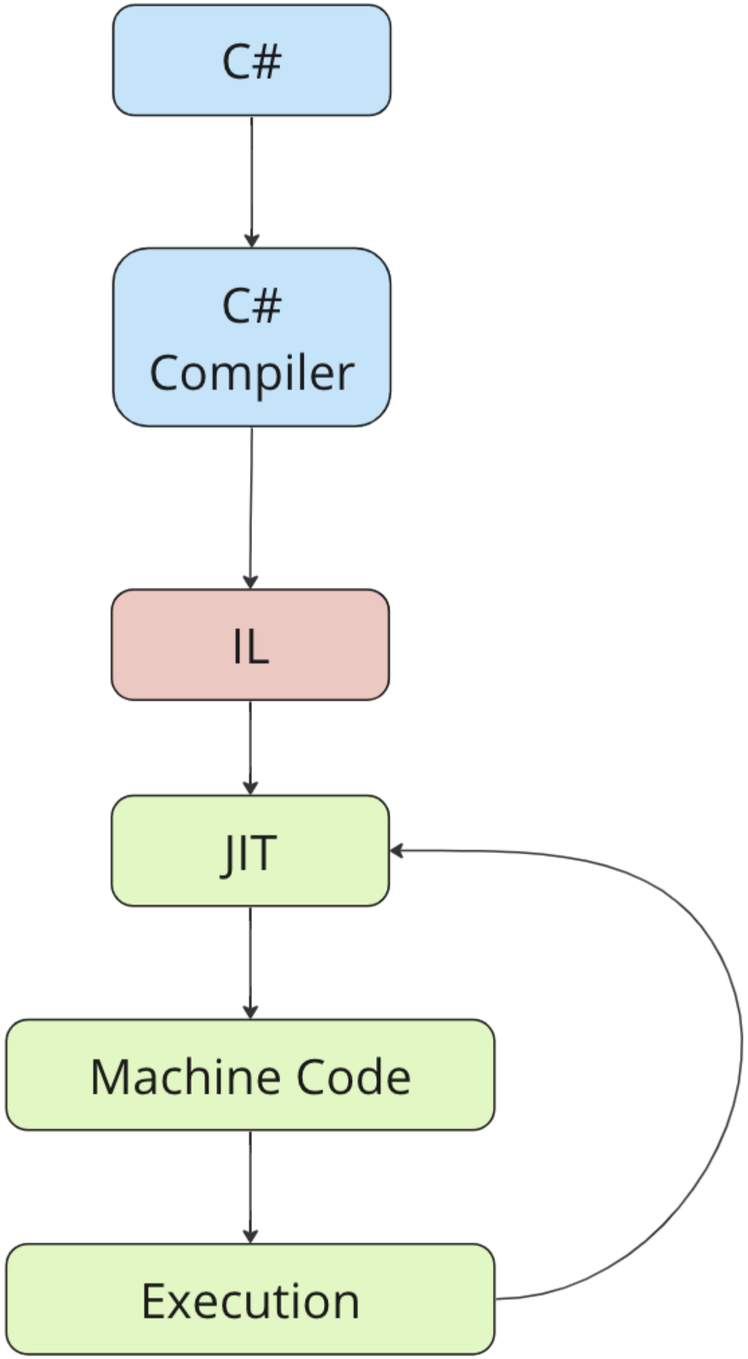
4. Conclusion

# Mono

Unity에서 기본적으로 사용되는 스크립팅 런타임

.NET Framework과 유사한 구조를 가지고 있으며,
C# 스크립트를 IL 코드로 변환하고,
JIT컴파일러를 사용하여 IL 코드를 실행

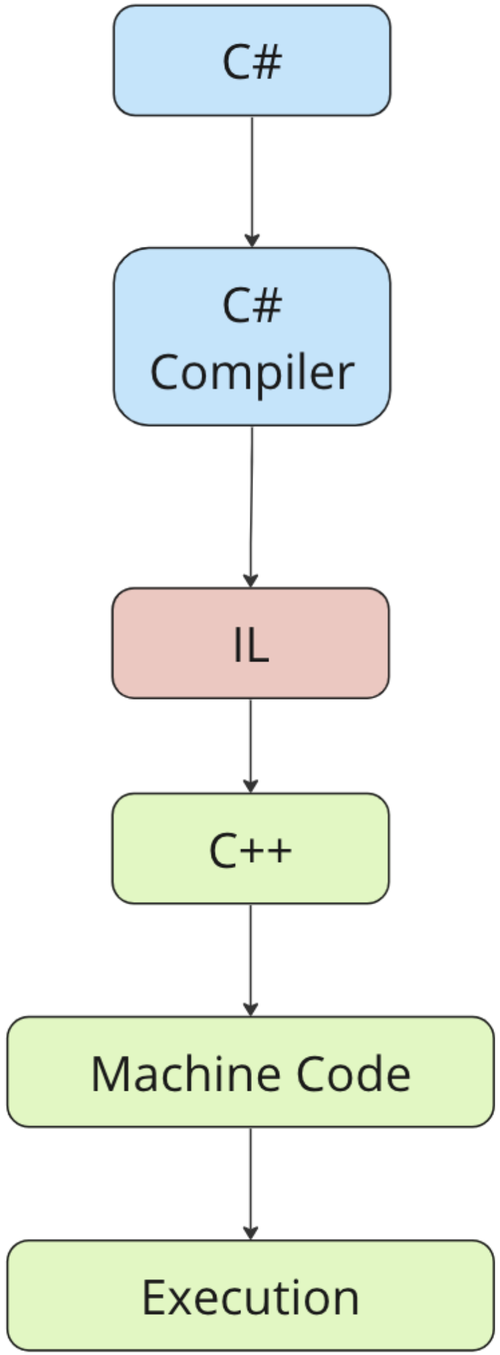
다양한 플랫폼에서 실행할 수 있고 빌드가 빠르지만..
DLL 파일을 통해 쉽게 디컴파일이 가능하여 보안에 취약하다.



# il2cpp

AOT(Ahead-Of-Time) 컴파일 방식을 따른다.
IL2CPP는 C# 코드를 컴파일하여 생성된 IL 코드를
C++ 코드로 변환한다.

런타임 시점에 코드를 컴파일하는 JIT 컴파일러와 달리
미리 코드를 컴파일하여 사용하기 때문에 보안에 강하다.



# il2cpp

하지만.. il2cpp 게임에는 Assembly-CSharp.dll 이 없다!!

| | | | |
|---|---|---|---|
| il2cpp_data | 2025-11-21 오후 1:43 | 파일 폴더 | |
| Plugins | 2025-11-21 오후 1:43 | 파일 폴더 | |
| Resources | 2025-11-21 오후 1:43 | 파일 폴더 | |
| StreamingAssets | 2025-11-21 오후 1:43 | 파일 폴더 | |
| app.info | 2025-11-17 오후 1:44 | INFO 파일 | 1KB |
| boot.config | 2025-11-21 오후 1:43 | XML Configuratio... | 1KB |
| data.unity3d | 2025-11-21 오후 1:43 | UNITY3D 파일 | 710,381KB |
| resources.resource | 2025-11-17 오후 1:44 | RESOURCE 파일 | 133KB |
| RuntimeInitializeOnLoads.json | 2025-11-17 오후 1:44 | JSON File | 9KB |
| ScriptingAssemblies.json | 2025-11-21 오후 1:43 | JSON File | 7KB |
| sharedassets1.resource | 2025-11-17 오후 1:44 | RESOURCE 파일 | 870KB |
| sharedassets5.resource | 2025-11-21 오후 1:43 | RESOURCE 파일 | 276KB |
| sharedassets6.resource | 2025-11-21 오후 1:43 | RESOURCE 파일 | 12,389KB |

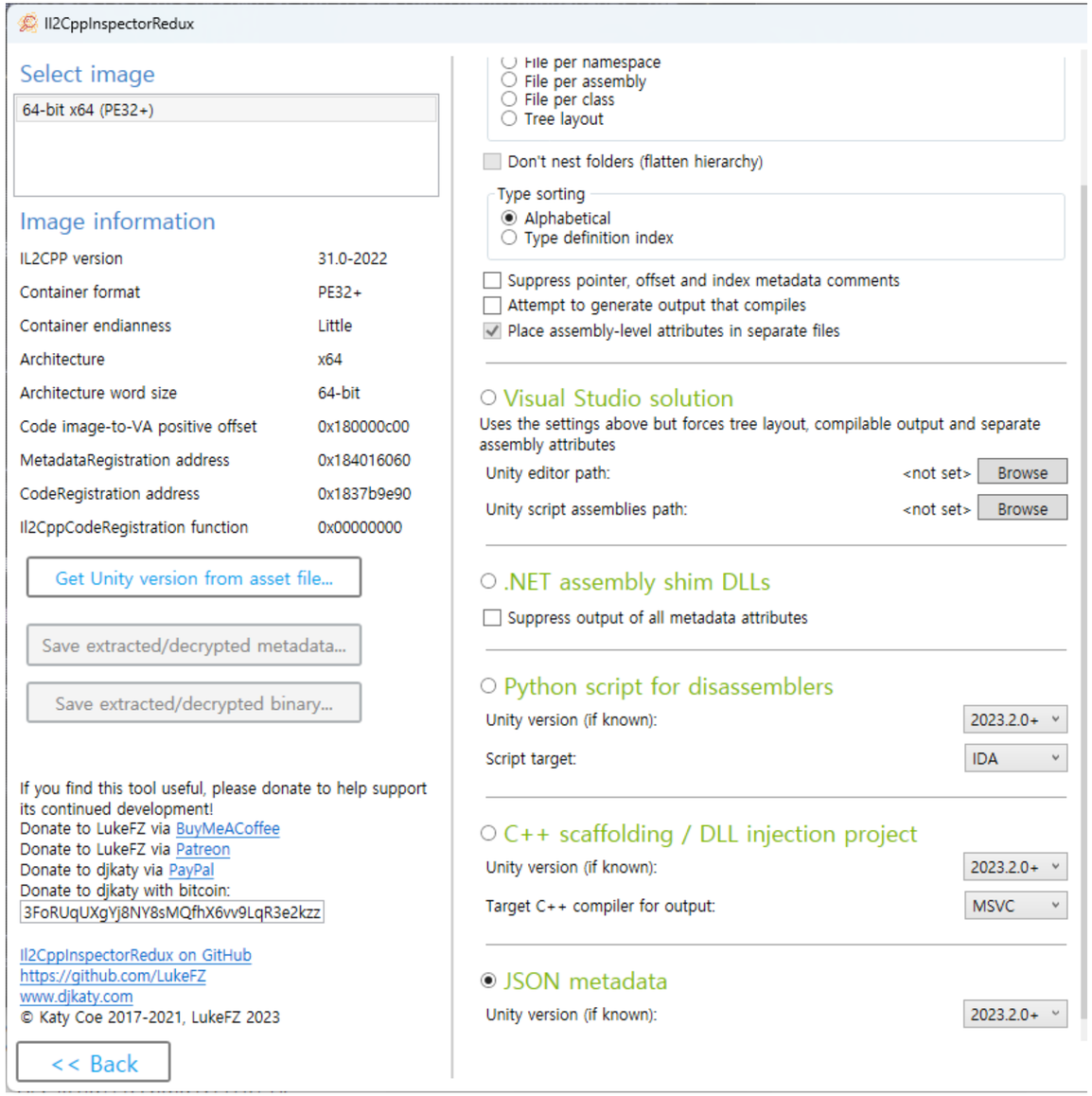
# il2cppinspector

그러나.. 이걸 추출해주는 툴이 있다..!!!

.NET assembly shim DLL 형태 (assembly-CSharp.dll 등)

Python script for disassemblers (함수 rename script)

Json metadata



# DnSpy

But.. 확인해보니 함수의 원본 내용은 확인할 수가 없다...!!!!!

```
// AnimatorRandomIdleSelector
// Token: 0x06000270 RID: 624 RVA: 0x000F8D4F File Offset: 0x000F6F4F
[Token(Token = "0x0600026E")]
[Address(RVA = "0x004AAE80", Offset = "0x4A9A80", VA = "0x00000001804AAE80", Slot = "4")]
public override void OnStateEnter(Animator animator, AnimatorStateInfo stateInfo, int layerIndex)
{
}
```

# DnSpy

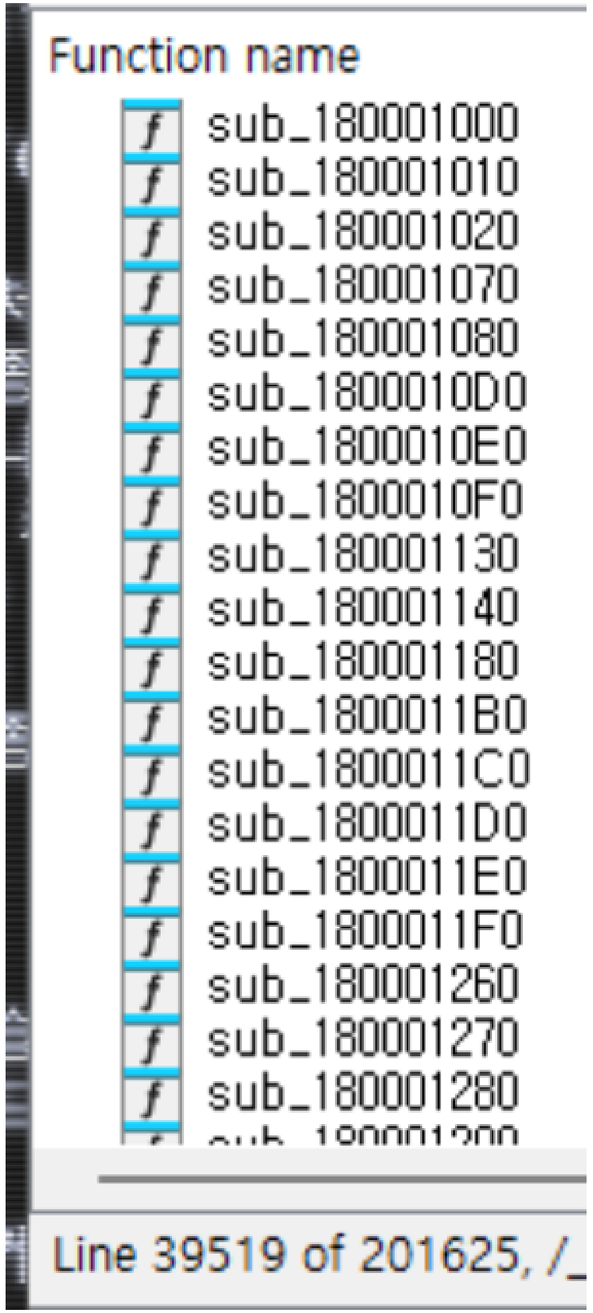
However.. RVA,VA 값을 확인할 수 있다!!!

-> IDA에서 해당 offset 함수 찾아 디컴파일 후 분석 진행

```
// AnimatorRandomIdleSelector
// Token: 0x06000270 RID: 624 RVA: 0x000F8D4F File Offset: 0x000F6F4F
[Token(Token = "0x0600026E")]
[Address(RVA = "0x004AAE80", Offset = "0x4A9A80", VA = "0x00000001804AAE80", Slot = "4")]
public override void OnStateEnter(Animator animator, AnimatorStateInfo stateInfo, int layerIndex)
{
}
```

# IDA

Nevertheless.. 함수의 개수가 말이 안된다...!!



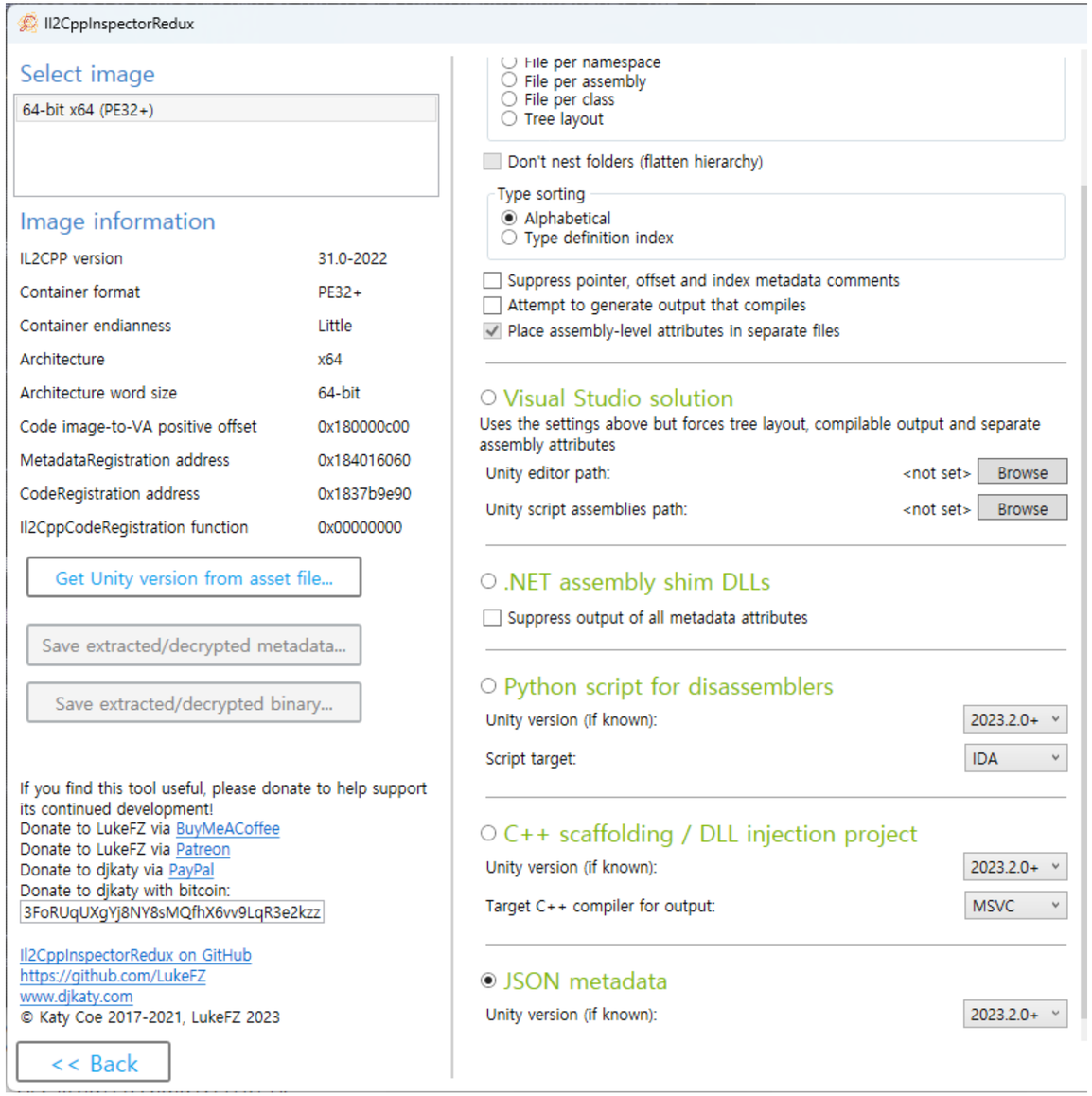
# il2cppinspector

그러나.. 이걸 추출해주는 툴이 있다..!!!

.NET assembly shim DLL 형태 (assembly-CSharp.dll 등)

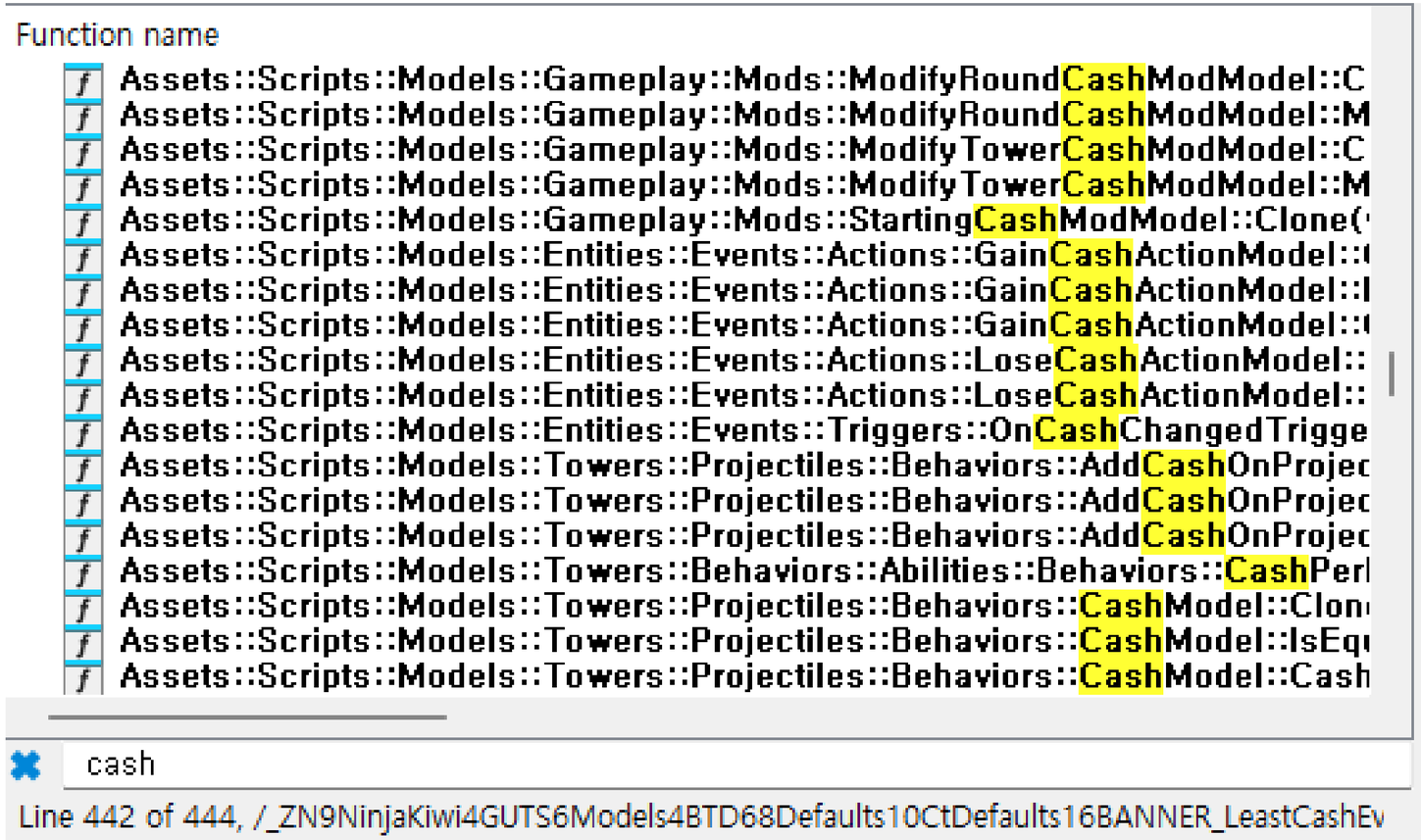
Python script for disassemblers (함수 rename script)

Json metadata



# IDA

그럼에도... 아직도 분석할 게 산 넘어 산이다..!!!



# IDA-pro-mcp

with Gemini Cli

```
✦ metadata.json 파일이 너무 커서 한 번에 읽을 수 없습니다.

  대신 파일 내용에서 "cash"와 관련된 키워드를 직접 검색하여 관련 함수를 찾겠습니다. "cash", "gold", "coin", "money",
  "currency", "reward" 순서로 검색을 진행하겠습니다.

  먼저 "cash"로 검색합니다.

  ✓  SearchText 'cash' within ./

  Found 2971 matches

✦ "cash" 검색 결과, 수많은 함수를 찾았습니다. 그중에서도 아래 함수들이 특히 인게임 캐시 획득과 관련이 깊어 보입니다:

  - Assets.Scripts.Simulation.Simulation.AddCash: 캐시를 추가하는 가장 기본적인 함수로 보입니다.
  - Assets.Scripts.Simulation.Simulation.AddSharedCash: 공유 캐시를 추가하는 함수입니다.
  - Assets.Scripts.Simulation.Towers.Projectiles.Behaviors.Cash.Pickup: 타워가 캐시를 획득하는 동작으로 추정됩니다.
  - Assets.Scripts.Simulation.Powers.CashDrop.Activate: '캐시 드롭' 파워를 활성화하는 함수입니다.
  - Assets.Scripts.Unity.UI_New.InGame.InGame.CheatSetCash: 이름으로 보아 캐시 치트와 관련된 함수일 가능성이
    높습니다.
  - Assets.Scripts.Behaviors.Events.Actions.GainCashAction.PerformAction: 캐시 획득 액션을 수행하는 함수입니다.

  이 중에서 가장 핵심적인 역할을 할 것으로 보이는 Assets.Scripts.Simulation.Simulation.AddCash 함수를 IDA에서 찾아
  분석해 보겠습니다. 함수의 Mangled Name은
  _ZN6Assets7Scripts10Simulation10Simulation7AddCashEdN6Assets7Scripts10Simulation10Simulation8CashTypeEiN6Assets7Scri
  pts10Simulation10Simulation10CashSourceEN6Assets7Scripts10Simulation6Towers5TowerEb 입니다.
```

# IDA-pro-mcp

with Gemini Cli

✦ AddCash 함수를 호출하는 다른 함수들에 대한 분석이 완료되었습니다. 이 함수들은 게임 내에서 캐시를 얻는 다양한 방법들을 보여줍니다.

주요 캐시 획득 경로:

* `AddSharedCash` / `SendCash`: 동료 플레이어와 캐시를 공유하거나 전송하는 기능 (멀티플레이)
* `SellTower`: 타워를 판매하여 일정량의 캐시를 회수
* `AddCashOnProjectileExpire` / `Cash.Exhausted` / `Cash.Pickup`: 특정 발사체나 오브젝트가 소멸되거나 획득될 때 캐시를 지급
* `Bank.Collect`: '바나나 농장'의 '은행' 업그레이드처럼, 저장된 캐시를 수집
* `PerRoundCashBonusTower.OnRoundComplete`: 라운드 종료 시 보너스 캐시를 제공하는 타워
* `CashPerBananaFarmInRange.Activate`: 범위 내의 바나나 농장 수에 따라 캐시를 획득하는 능력
* `ImfLoan.Activate`: 'IMF 대출' 능력처럼 큰 금액의 캐시를 즉시 받고 나중에 상환하는 기능
* `PropManager.SellProp`: 맵 에디터에서 장식(prop)을 판매하여 캐시 획득
* `Nourishment.Cast`: 캐시를 제공하는 특정 주문 또는 능력
* `EndOfRoundCashBoostBehavior.OnRoundEnd`: 라운드 종료 시 캐시를 증폭시키는 효과

분석 요약:

캐시는 게임 내에서 매우 다양한 방법으로 획득할 수 있습니다. 가장 일반적인 방법은 '바나나 농장'과 같은 캐시 생성 타워를 이용하는 것이지만, 타워 판매, 특수 능력, 라운드 보너스, 멀티플레이 상호작용 등 여러 경로가 존재합니다.

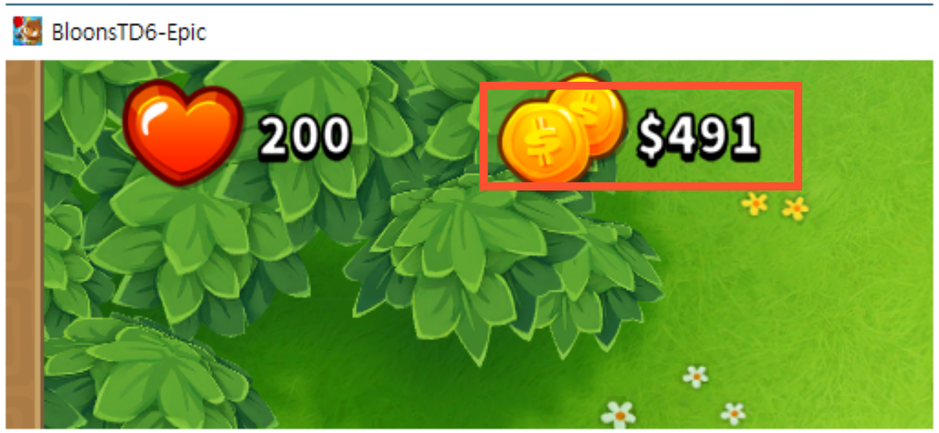
AddCash 함수와 이를 호출하는 함수들의 분석을 통해 인게임 캐시 획득 방법에 대한 종합적인 분석을 마쳤습니다.

# Game Analyze

Bloons TD6

# Game Analyze

372원 상태에서 119원을 팔아 491원이 됨




491 - 372 =

119

# Cheat Engine

Cheat Engine 에서 캐시값 찾아내고, 해당 변수 호출하는 함수를 찾았으나..

| | | | | |
|---|---|---|---|---|
| GameAssembly.dll+B3CE20 | 40 53 | push | rbx | |
| GameAssembly.dll+B3CE22 | 48 83 EC 30 | sub | rsp,30 | 48 |
| GameAssembly.dll+B3CE26 | 0F29 74 24 20 | movaps | [rsp+20],xmm6 | |
| GameAssembly.dll+B3CE2B | 45 33 C0 | xor | r8d,r8d | |
| GameAssembly.dll+B3CE2E | 0F28 F1 | movaps | xmm6,xmm1 | |
| GameAssembly.dll+B3CE31 | 48 8B D9 | mov | rbx,rcx | |
| GameAssembly.dll+B3CE34 | E8 B7FDFFFF | call | GameAssembly.dll+B3CBF0 | |
| GameAssembly.dll+B3CE39 | 48 8B 43 28 | mov | rax,[rbx+28] | |
| GameAssembly.dll+B3CE3D | F2 0F11 73 28 | movsd | [rbx+28],xmm6 | |
| GameAssembly.dll+B3CE42 | 0F28 74 24 20 | movaps | xmm6,[rsp+20] | |
| GameAssembly.dll+B3CE47 | 48 89 43 30 | mov | [rbx+30],rax | |
| GameAssembly.dll+B3CE4B | 48 83 C4 30 | add | rsp,30 | 48 |
| GameAssembly.dll+B3CE4F | 5B | pop | rbx | |

# Cheat Engine

Cheat Engine 에서 캐시값 찾아내고, 해당 변수 호출하는 함수를 찾았으나..

```
Registers:
RAX 4077400000000000
RBX 00000280519E7A50
RCX 000000007A58D853
RDX 000000000009D757
RSI 00002803F6C7640
RDI 00000280051570E00
```

Double-precision floating point

Hex value: 0x4077400000000000 Convert to double

0x4077400000000000

| 4 | | | | 0 | | | | 7 | | | | 7 | | | | 4 | | | | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 10000000111 | | | | | | | | | | | 0111010000000000000000 | | | | | | | | | | | |

| sign | exponent | mantissa |
|---|---|---|
| +1 | 1031 | 1.0111010000000000000000 |
| +1 * | 2^(1031 - 1023) * | 1.453125 |
| +1 * | 256.00000000000000 * | 1.453125 |

372

Double value: 372 Convert to hex

# IDA

Cheat Engine 에서 캐시값 찾아내고, 해당 변수 호출하는 함수를 찾았으나..

```
GameAssembly.dll+B3CE20   40 53          push    rbx
GameAssembly.dll+B3CE22   48 83 EC 30    sub     rsp,30                    48
GameAssembly.dll+B3CE26   0F29 74 24 20  movaps  [rsp+20],xmm6
GameAssembly.dll+B3CE2B   45 33 C0       xor     r8d,r8d
GameAssembly.dll+B3CE2E   0F28 F1        movaps  xmm6,xmm1
GameAssembly.dll+B3CE31   48 8B D9       mov     rbx,rcx
GameAssembly.dll+B3CE34   E8 B7FDFFFF    call    GameAssembly.dll+B3CBF0
GameAssembly.dll+B3CE39   48 8B 43 28    mov     rax,[rbx+28]
GameAssembly.dll+B3CE3D   F2 0F11 73 28  movsd   [rbx+28],xmm6
GameAssembly.dll+B3CE42
GameAssembly.dll+B3CE47
GameAssembly.dll+B3CE4B
GameAssembly.dll+B3CE4F
```

```
// Void Write(Double)
void Assets::Scripts::Utils::KonFuze::Write(KonFuze *this, double v, MethodInfo *method)
{
  double honey; // rax

  Assets::Scripts::Utils::KonFuze::WriteRemoved(this, v, 0LL);
  honey = this->fields.honey;
  this->fields.honey = v;
  this->fields.pot = honey;
}
```

# IDA

해당 Write 함수를 호출하여 넘겨 넘겨 값을 변경하는 방식... 계속해서 호출이 진행됨

xrefs to Assets::Scripts::Utils::KonFuze::Write(double)

| Directic | Type | Address | Text | |
|---|---|---|---|---|
| Up | p | RogueExtractionPanel::_ExtractArtifact_d__24::MoveNext(void)+4A0 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnAbilityActivated(Assets::Scripts::Simulation::Towers::Tower,Assets::Scr··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnBADPop(Assets::Scripts::Simulation::Towers::Tower)+AD | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnBFBPop(Assets::Scripts::Simulation::Towers::Tower)+AA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnBossDamage(Assets::Scripts::Simulation::Towers::Tower,long)+B6 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnBossPop(Assets::Scripts::Simulation::Towers::Tower)+AD | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnCamoPop(int,Assets::Scripts::Simulation::Towers::Tower)+B2 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnCashEarned(int,double,Assets::Scripts::Simulation::Simulation::CashTy··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnCeramicPop(Assets::Scripts::Simulation::Towers::Tower)+AA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnDDTPop(Assets::Scripts::Simulation::Towers::Tower)+AD | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnLeadPop(Assets::Scripts::Simulation::Towers::Tower)+AA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnMOABPop(Assets::Scripts::Simulation::Towers::Tower)+AA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnParagonUpgraded(Assets::Scripts::Simulation::Towers::Tower,float,dou··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnPopBloon(int,Assets::Scripts::Simulation::Towers::Tower)+F3 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnPurplePop(Assets::Scripts::Simulation::Towers::Tower)+AA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnRegrowPop(int,Assets::Scripts::Simulation::Towers::Tower)+B2 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnRoundEnd(int)+4CA | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnTowerPlaced(Assets::Scripts::Simulation::Towers::Tower,Assets::Scrip··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnTowerPlaced(Assets::Scripts::Simulation::Towers::Tower,Assets::Scrip··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnTowerSacrificed(Assets::Scripts::Simulation::Towers::Tower,Assets::S··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnTowerUpgraded(Assets::Scripts::Simulation::Towers::Tower,Assets::Sc··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnWon(void)+25A | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Tracking::AnalyticsTrackerSim::OnZOMGPop(Assets::Scripts::Simulation::Towers::Tower)+AD | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::AddCash(double,Assets::Scripts::Simulation::Simulation::CashType,int,Assets::Scripts::Simula··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::DeductSharedCash(double,Assets::Scripts::Simulation::Simulation::CashType,Assets::Scripts:··· | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+E0 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+104 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+124 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+144 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+168 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+188 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+1B8 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+1D8 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+270 | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::Initialise(Assets::Scripts::Models::GameModel,bool)+28B | call | Assets::Scripts::Utils::KonFuze::Wi |
| Up | p | Assets::Scripts::Simulation::Simulation::RemoveCash(double,Assets::Scripts::Simulation::Simulation::CashType,int,Assets::Scripts::Si··· | call | Assets::Scripts::Utils::KonFuze::Wi |

Line 24 of 485

OK Cancel Search Help

■ ■ ■

```
// Double AddCash(Double, Simulation+CashType, Int32, Simulation+CashSource, Tower, Boolean)
double Assets::Scripts::Simulation::Simulation::AddCash(
        Simulation *this,
        double c,
        Simulation_CashType__Enum from,
        int32_t cashIndex,
        Simulation_CashSource__Enum source,
        Tower *tower,
        bool cashDoubleable,
        MethodInfo *method)
{
  List_1_Assets_Scripts_Simulation_SimulationBehaviors_SimulationBehavior_ *behaviorCache; // rbp
  char v13; // r14
  int size; // ebp
```

```
GameAssembly.dll+4C7F30   40 56                 push   rsi
GameAssembly.dll+4C7F32   41 54                 push   r12
GameAssembly.dll+4C7F34   41 55                 push   r13
GameAssembly.dll+4C7F36   41 57                 push   r15
GameAssembly.dll+4C7F38   48 83 EC 58           sub    rsp,58                          88
GameAssembly.dll+4C7F3C   80 3D 22624404 00     cmp    byte ptr [GameAssembly.dll+490E1(1),0
GameAssembly.dll+4C7F43   45 8B F9              mov    r15d,r9d
GameAssembly.dll+4C7F46   0F29 74 24 40         movaps [rsp+40],xmm6
GameAssembly.dll+4C7F4B   45 8B E0              mov    r12d,r8d
GameAssembly.dll+4C7F4E   0F28 F1               movaps xmm6,xmm1
GameAssembly.dll+4C7F51   48 8B F1              mov    rsi,rcx
GameAssembly.dll+4C7F54   75 37                 jne    GameAssembly.dll+4C7F8D
GameAssembly.dll+4C7F56   48 8D 0D C3BB1D04     lea    rcx,[GameAssembly.dll+46A3B20](27E2187B780)
GameAssembly.dll+4C7F5D   E8 7E9FEEFF           call   GameAssembly.il2cpp_get_except
GameAssembly.dll+4C7F62   48 8D 0D F7982104     lea    rcx,[GameAssembly.dll+46E1860](27E2729A6E8)
GameAssembly.dll+4C7F69   E8 729FEEFF           call   GameAssembly.il2cpp_get_except
GameAssembly.dll+4C7F6E   48 8D 0D 0B9B2104     lea    rcx,[GameAssembly.dll+46E1A80](27E272990D0)
GameAssembly.dll+4C7F75   E8 669FEEFF           call   GameAssembly.il2cpp_get_except
GameAssembly.dll+4C7F7A   48 8D 0D 0F9C2104     lea    rcx,[GameAssembly.dll+46E1B90](27E27299378)
GameAssembly.dll+4C7F81   E8 5A9FEEFF           call   GameAssembly.il2cpp_get_except
GameAssembly.dll+4C7F86   C6 05 D8614404 01     mov    byte ptr [GameAssembly.dll+490E1(1),1
GameAssembly.dll+4C7F8D   44 8B AC 24 A0000000  mov    r13d,[rsp+000000A0]
GameAssembly.dll+4C7F95   48 89 9C 24 80000000  mov    [rsp+00000080],rbx
GameAssembly.dll+4C7F9D   48 89 AC 24 88000000  mov    [rsp+00000088],rbp
GameAssembly.dll+4C7FA5   48 89 BC 24 90000000  mov    [rsp+00000090],rdi
GameAssembly.dll+4C7FAD   4C 89 74 24 50        mov    [rsp+50],r14
GameAssembly.dll+4C7FB2   41 83 FC 04           cmp    r12d,04                         4
GameAssembly.dll+4C7FB6   0F84 DE000000         je     GameAssembly.dll+4C809A
GameAssembly.dll+4C7FBC   48 8B AE D8010000     mov    rbp,[rsi+000001D8]
GameAssembly.dll+4C7FC3   45 32 F6              xor    r14b,r14b
GameAssembly.dll+4C7FC6   48 85 ED              test   rbp,rbp
GameAssembly.dll+4C7FC9   0F84 C3010000         je     GameAssembly.dll+4C8192
GameAssembly.dll+4C7FCF   8B 6D 18              mov    ebp,[rbp+18]
GameAssembly.dll+4C7FD2   33 DB                 xor    ebx,ebx
GameAssembly.dll+4C7FD4   85 ED                 test   ebp,ebp
```

```
tionBehavior_ *v16; // rcx

rcx

ripts::Simulation::SimulationBehaviors::ImfLoanCollection);
Collections::Generic::List<Assets::Scripts::Simulation::SimulationBehaviors::SimulationBehavior>::Remov
Collections::Generic::List<Assets::Scripts::Simulation::SimulationBehaviors::SimulationBehavior>::get_(
Collections::Generic::List<Assets::Scripts::Simulation::SimulationBehaviors::SimulationBehavior>::get_
```

## Game Analyze

# ! ! !

```
E8 1D7B0600      call    GameAssembly.dll+52FC30
F2 0F58 C6       addsd   xmm0,xmm6
```

FPU

Extended

XMMRegisters | Classic view

Double

| | | |
|---|---|---|
| XMM0: | 429,00 | 0,00 |
| XMM1: | 429,00 | 0,00 |
| XMM2: | 0,00 | 0,00 |
| XMM3: | 0,00 | 0,00 |
| XMM4: | 0,00 | 0,00 |
| XMM5: | 0,00 | 0,00 |
| XMM6: | 999999,00 | 0,00 |
| XMM7: | 0,00 | 0,00 |
| XMM8: | 429,00 | 0,00 |
| XMM9: | 0,00 | 0,00 |
| XMM10: | 0,00 | 0,00 |
| XMM11: | 0,00 | 0,00 |
| XMM12: | 0,00 | 0,00 |
| XMM13: | 0,00 | 0,00 |
| XMM14: | 0,00 | 0,00 |
| XMM15: | 0,00 | 0,00 |

# !!!

$1,000,428

# Conclusion

바이너리 크기가 크기인 만큼, 시간과 오류가 동반한다..
-> AssultCube 같은 오픈소스 게임은 분석이 용이했으나 실제 서비스중인 게임은 그 기능이 훨씬 방대하다

결국 마지막 동적 분석의 영역에서는 아직 리버싱의 역량이 필요하다..

+ 모바일도 전부 똑같이 진행 가능하고, Frida hook script 까지 만들어주지만 루팅된 폰이 없음....
(Nox 등의 에뮬레이터는 ARM 환경과 달라 올바른 메모리 스캐닝이 안됨)

# Q & A